

8

8.2 Payload

In the **Payload** tab, you can set the weight to be handled by the end of the robotic arm. The robot can achieve optimal performance with a proper setting of the Payload parameters.

The screenshot shows the DRASudio software interface with the 'Payload' tab selected. The 'Menu' sidebar on the left contains an 'Arm' icon. The main area displays the following parameters and controls:

- Model:** A dropdown menu showing 'DRV70L7A631' with 'Get' and 'Set' buttons.
- Mass:** A text input field containing '4' followed by 'kg'.
- Distance:** Three text input fields for 'CoGx' (0), 'CoGy' (0), and 'CoGz' (0.092) followed by 'm'.
- Inertia:** Six text input fields for 'ixx' (0.019), 'iyy' (0.025), 'izz' (0.017), 'ixy' (0), 'ixz' (0), and 'iyz' (0) followed by 'kg*m^2'.
- J5 Inertia:** A text input field followed by 'kg*m^2'.
- J6 Inertia:** A text input field followed by 'kg*m^2'.
- Calculate:** A button to calculate the inertia values.

Setting steps:

1. Click  (Servo control) in the Quick access toolbar to switch all the motors to the Servo Off state.
2. Properly set the Mass, Distance, and Inertia.
3. Click **Set**.

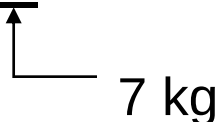
Important: pay special attention when setting the parameters in the **Payload** tab, which have significant impact on the robot's moving speed.

8.2.1 Setting the Mass

Set the weight (unit: kg) to be handled by the end of the robotic arm. If you set the weight too high, the robot handles the end effector with higher torque, which leads to inefficiencies. If you set the weight too low, the robot may jitter because of insufficient torque.

The robot's payload capacity differs from models. For example, in the model number of DRV70L7, the last code "7" indicates a payload of 7 kg (15.4 lb).

DRV70L7



7 kg